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*To Suzanne, Barbara, Daniel, Aron, Marvin, Matilde,
and the memory of Bram, and Sweetie π (AST)*

To Katrin, Lucy, and Pepper (DJW)

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CONTENTS

PREFACE

xix

1 INTRODUCTION

1

1.1 USES OF COMPUTER NETWORKS, 3

1.1.1 Business Applications, 3

1.1.2 Home Applications, 6

1.1.3 Mobile Users, 10

1.1.4 Social Issues, 14

1.2 NETWORK HARDWARE, 17

1.2.1 Personal Area Networks, 18

1.2.2 Local Area Networks, 19

1.2.3 Metropolitan Area Networks, 23

1.2.4 Wide Area Networks, 23

1.2.5 Internetworks, 28

1.3 NETWORK SOFTWARE, 29

1.3.1 Protocol Hierarchies, 29

1.3.2 Design Issues for the Layers, 33

1.3.3 Connection-Oriented Versus Connectionless Service, 35

1.3.4 Service Primitives, 38

1.3.5 The Relationship of Services to Protocols, 40

1.4 REFERENCE MODELS, 41

1.4.1 The OSI Reference Model, 41

1.4.2 The TCP/IP Reference Model, 45

1.4.3 The Model Used in This Book, 48

- 1.4.4 A Comparison of the OSI and TCP/IP Reference Models*, 49
- 1.4.5 A Critique of the OSI Model and Protocols*, 51
- 1.4.6 A Critique of the TCP/IP Reference Model*, 53
- 1.5 EXAMPLE NETWORKS, 54
 - 1.5.1 The Internet, 54
 - 1.5.2 Third-Generation Mobile Phone Networks*, 65
 - 1.5.3 Wireless LANs: 802.11*, 70
 - 1.5.4 RFID and Sensor Networks*, 73
- 1.6 NETWORK STANDARDIZATION*, 75
 - 1.6.1 Who's Who in the Telecommunications World, 77
 - 1.6.2 Who's Who in the International Standards World, 78
 - 1.6.3 Who's Who in the Internet Standards World, 80
- 1.7 METRIC UNITS, 82
- 1.8 OUTLINE OF THE REST OF THE BOOK, 83
- 1.9 SUMMARY, 84

2 THE PHYSICAL LAYER

89

- 2.1 THE THEORETICAL BASIS FOR DATA COMMUNICATION, 90
 - 2.1.1 Fourier Analysis, 90
 - 2.1.2 Bandwidth-Limited Signals, 90
 - 2.1.3 The Maximum Data Rate of a Channel, 94
- 2.2 GUIDED TRANSMISSION MEDIA, 95
 - 2.2.1 Magnetic Media, 95
 - 2.2.2 Twisted Pairs, 96
 - 2.2.3 Coaxial Cable, 97
 - 2.2.4 Power Lines, 98
 - 2.2.5 Fiber Optics, 99
- 2.3 WIRELESS TRANSMISSION, 105
 - 2.3.1 The Electromagnetic Spectrum, 105
 - 2.3.2 Radio Transmission, 109
 - 2.3.3 Microwave Transmission, 110
 - 2.3.4 Infrared Transmission, 114
 - 2.3.5 Light Transmission, 114

- 2.4 COMMUNICATION SATELLITES*, 116
 - 2.4.1 Geostationary Satellites, 117
 - 2.4.2 Medium-Earth Orbit Satellites, 121
 - 2.4.3 Low-Earth Orbit Satellites, 121
 - 2.4.4 Satellites Versus Fiber, 123
- 2.5 DIGITAL MODULATION AND MULTIPLEXING, 125
 - 2.5.1 Baseband Transmission, 125
 - 2.5.2 Passband Transmission, 130
 - 2.5.3 Frequency Division Multiplexing, 132
 - 2.5.4 Time Division Multiplexing, 135
 - 2.5.5 Code Division Multiplexing, 135
- 2.6 THE PUBLIC SWITCHED TELEPHONE NETWORK, 138
 - 2.6.1 Structure of the Telephone System, 139
 - 2.6.2 The Politics of Telephones, 142
 - 2.6.3 The Local Loop: Modems, ADSL, and Fiber, 144
 - 2.6.4 Trunks and Multiplexing, 152
 - 2.6.5 Switching, 161
- 2.7 THE MOBILE TELEPHONE SYSTEM*, 164
 - 2.7.1 First-Generation (coco1G) Mobile Phones: Analog Voice, 166
 - 2.7.2 Second-Generation (2G) Mobile Phones: Digital Voice, 170
 - 2.7.3 Third-Generation (3G) Mobile Phones: Digital Voice and Data, 174
- 2.8 CABLE TELEVISION*, 179
 - 2.8.1 Community Antenna Television, 179
 - 2.8.2 Internet over Cable, 180
 - 2.8.3 Spectrum Allocation, 182
 - 2.8.4 Cable Modems, 183
 - 2.8.5 ADSL Versus Cable, 185
- 2.9 SUMMARY, 186

3 THE DATA LINK LAYER

193

- 3.1 DATA LINK LAYER DESIGN ISSUES, 194
 - 3.1.1 Services Provided to the Network Layer, 194
 - 3.1.2 Framing, 197
 - 3.1.3 Error Control, 200
 - 3.1.4 Flow Control, 201

- 3.2 ERROR DETECTION AND CORRECTION, 202
 - 3.2.1 Error-Correcting Codes, 204
 - 3.2.2 Error-Detecting Codes, 209
- 3.3 ELEMENTARY DATA LINK PROTOCOLS, 215
 - 3.3.1 A Utopian Simplex Protocol, 220
 - 3.3.2 A Simplex Stop-and-Wait Protocol for an Error-Free Channel, 221
 - 3.3.3 A Simplex Stop-and-Wait Protocol for a Noisy Channel, 222
- 3.4 SLIDING WINDOW PROTOCOLS, 226
 - 3.4.1 A One-Bit Sliding Window Protocol, 229
 - 3.4.2 A Protocol Using Go-Back-N, 232
 - 3.4.3 A Protocol Using Selective Repeat, 239
- 3.5 EXAMPLE DATA LINK PROTOCOLS, 244
 - 3.5.1 Packet over SONET, 245
 - 3.5.2 ADSL (Asymmetric Digital Subscriber Loop), 248
- 3.6 SUMMARY, 251

4 THE MEDIUM ACCESS CONTROL SUBLAYER 257

- 4.1 THE CHANNEL ALLOCATION PROBLEM, 258
 - 4.1.1 Static Channel Allocation, 258
 - 4.1.2 Assumptions for Dynamic Channel Allocation, 260
- 4.2 MULTIPLE ACCESS PROTOCOLS, 261
 - 4.2.1 ALOHA, 262
 - 4.2.2 Carrier Sense Multiple Access Protocols, 266
 - 4.2.3 Collision-Free Protocols, 269
 - 4.2.4 Limited-Contention Protocols, 274
 - 4.2.5 Wireless LAN Protocols, 277
- 4.3 ETHERNET, 280
 - 4.3.1 Classic Ethernet Physical Layer, 281
 - 4.3.2 Classic Ethernet MAC Sublayer Protocol, 282
 - 4.3.3 Ethernet Performance, 286
 - 4.3.4 Switched Ethernet, 288

- 4.3.5 Fast Ethernet, 290
- 4.3.6 Gigabit Ethernet, 293
- 4.3.7 10-Gigabit Ethernet, 296
- 4.3.8 Retrospective on Ethernet, 298
- 4.4 WIRELESS LANS, 299
 - 4.4.1 The 802.11 Architecture and Protocol Stack, 299
 - 4.4.2 The 802.11 Physical Layer, 301
 - 4.4.3 The 802.11 MAC Sublayer Protocol, 303
 - 4.4.4 The 802.11 Frame Structure, 309
 - 4.4.5 Services, 311
- 4.5 BROADBAND WIRELESS*, 312
 - 4.5.1 Comparison of 802.16 with 802.11 and 3G, 313
 - 4.5.2 The 802.16 Architecture and Protocol Stack, 314
 - 4.5.3 The 802.16 Physical Layer, 316
 - 4.5.4 The 802.16 MAC Sublayer Protocol, 317
 - 4.5.5 The 802.16 Frame Structure, 319
- 4.6 BLUETOOTH*, 320
 - 4.6.1 Bluetooth Architecture, 320
 - 4.6.2 Bluetooth Applications, 321
 - 4.6.3 The Bluetooth Protocol Stack, 322
 - 4.6.4 The Bluetooth Radio Layer, 324
 - 4.6.5 The Bluetooth Link Layers, 324
 - 4.6.6 The Bluetooth Frame Structure, 325
- 4.7 RFID*, 327
 - 4.7.1 EPC Gen 2 Architecture, 327
 - 4.7.2 EPC Gen 2 Physical Layer, 328
 - 4.7.3 EPC Gen 2 Tag Identification Layer, 329
 - 4.7.4 Tag Identification Message Formats, 331
- 4.8 DATA LINK LAYER SWITCHING, 332
 - 4.8.1 Uses of Bridges, 332
 - 4.8.2 Learning Bridges, 334
 - 4.8.3 Spanning Tree Bridges, 337
 - 4.8.4 Repeaters, Hubs, Bridges, Switches, Routers, and Gateways, 340
 - 4.8.5 Virtual LANs, 342
- 4.9 SUMMARY, 349

5 THE NETWORK LAYER

355

5.1 NETWORK LAYER DESIGN ISSUES, 355

- 5.1.1 Store-and-Forward Packet Switching, 356
- 5.1.2 Services Provided to the Transport Layer, 356
- 5.1.3 Implementation of Connectionless Service, 358
- 5.1.4 Implementation of Connection-Oriented Service, 359
- 5.1.5 Comparison of Virtual-Circuit and Datagram Networks, 361

5.2 ROUTING ALGORITHMS, 362

- 5.2.1 The Optimality Principle, 364
- 5.2.2 Shortest Path Algorithm, 366
- 5.2.3 Flooding, 368
- 5.2.4 Distance Vector Routing, 370
- 5.2.5 Link State Routing, 373
- 5.2.6 Hierarchical Routing, 378
- 5.2.7 Broadcast Routing, 380
- 5.2.8 Multicast Routing, 382
- 5.2.9 Anycast Routing, 385
- 5.2.10 Routing for Mobile Hosts, 386
- 5.2.11 Routing in Ad Hoc Networks, 389

5.3 CONGESTION CONTROL ALGORITHMS, 392

- 5.3.1 Approaches to Congestion Control, 394
- 5.3.2 Traffic-Aware Routing, 395
- 5.3.3 Admission Control, 397
- 5.3.4 Traffic Throttling, 398
- 5.3.5 Load Shedding, 401

5.4 QUALITY OF SERVICE, 404

- 5.4.1 Application Requirements, 405
- 5.4.2 Traffic Shaping, 407
- 5.4.3 Packet Scheduling, 411
- 5.4.4 Admission Control, 415
- 5.4.5 Integrated Services, 418
- 5.4.6 Differentiated Services, 421

5.5 INTERNETWORKING, 424

- 5.5.1 How Networks Differ, 425
- 5.5.2 How Networks Can Be Connected, 426
- 5.5.3 Tunneling, 429

5.5.4 Internetwork Routing, 431

5.5.5 Packet Fragmentation, 432

5.6 THE NETWORK LAYER IN THE INTERNET, 436

5.6.1 The IP Version 4 Protocol, 439

5.6.2 IP Addresses, 442

5.6.3 IP Version 6, 455

5.6.4 Internet Control Protocols, 465

5.6.5 Label Switching and MPLS, 470

5.6.6 OSPF—An Interior Gateway Routing Protocol, 474

5.6.7 BGP—The Exterior Gateway Routing Protocol, 479

5.6.8 Internet Multicasting, 484

5.6.9 Mobile IP, 485

5.7 SUMMARY, 488

6 THE TRANSPORT LAYER

495

6.1 THE TRANSPORT SERVICE, 495

6.1.1 Services Provided to the Upper Layers, 496

6.1.2 Transport Service Primitives, 498

6.1.3 Berkeley Sockets, 500

6.1.4 An Example of Socket Programming: An Internet File Server, 503

6.2 ELEMENTS OF TRANSPORT PROTOCOLS, 507

6.2.1 Addressing, 509

6.2.2 Connection Establishment, 512

6.2.3 Connection Release, 517

6.2.4 Error Control and Flow Control, 522

6.2.5 Multiplexing, 527

6.2.6 Crash Recovery, 527

6.3 CONGESTION CONTROL, 530

6.3.1 Desirable Bandwidth Allocation, 531

6.3.2 Regulating the Sending Rate, 535

6.3.3 Wireless Issues, 539

6.4 THE INTERNET TRANSPORT PROTOCOLS: UDP, 541

6.4.1 Introduction to UDP, 541

6.4.2 Remote Procedure Call, 543

6.4.3 Real-Time Transport Protocols, 546

6.5 THE INTERNET TRANSPORT PROTOCOLS: TCP, 552

- 6.5.1 Introduction to TCP, 552
- 6.5.2 The TCP Service Model, 553
- 6.5.3 The TCP Protocol, 556
- 6.5.4 The TCP Segment Header, 557
- 6.5.5 TCP Connection Establishment, 560
- 6.5.6 TCP Connection Release, 562
- 6.5.7 TCP Connection Management Modeling, 562
- 6.5.8 TCP Sliding Window, 565
- 6.5.9 TCP Timer Management, 568
- 6.5.10 TCP Congestion Control, 571
- 6.5.11 The Future of TCP, 581

6.6 PERFORMANCE ISSUES*, 582

- 6.6.1 Performance Problems in Computer Networks, 583
- 6.6.2 Network Performance Measurement, 584
- 6.6.3 Host Design for Fast Networks, 586
- 6.6.4 Fast Segment Processing, 590
- 6.6.5 Header Compression, 593
- 6.6.6 Protocols for Long Fat Networks, 595

6.7 DELAY-TOLERANT NETWORKING*, 599

- 6.7.1 DTN Architecture, 600
- 6.7.2 The Bundle Protocol, 603

6.8 SUMMARY, 605**7 THE APPLICATION LAYER****611****7.1 DNS—THE DOMAIN NAME SYSTEM, 611**

- 7.1.1 The DNS Name Space, 612
- 7.1.2 Domain Resource Records, 616
- 7.1.3 Name Servers, 619

7.2 ELECTRONIC MAIL*, 623

- 7.2.1 Architecture and Services, 624
- 7.2.2 The User Agent, 626
- 7.2.3 Message Formats, 630
- 7.2.4 Message Transfer, 637
- 7.2.5 Final Delivery, 643

- 7.3 THE WORLD WIDE WEB, 646
 - 7.3.1 Architectural Overview, 647
 - 7.3.2 Static Web Pages, 662
 - 7.3.3 Dynamic Web Pages and Web Applications, 672
 - 7.3.4 HTTP—The HyperText Transfer Protocol, 683
 - 7.3.5 The Mobile Web, 693
 - 7.3.6 Web Search, 695
- 7.4 STREAMING AUDIO AND VIDEO, 697
 - 7.4.1 Digital Audio, 699
 - 7.4.2 Digital Video, 704
 - 7.4.3 Streaming Stored Media, 713
 - 7.4.4 Streaming Live Media, 721
 - 7.4.5 Real-Time Conferencing, 724
- 7.5 CONTENT DELIVERY, 734
 - 7.5.1 Content and Internet Traffic, 736
 - 7.5.2 Server Farms and Web Proxies, 738
 - 7.5.3 Content Delivery Networks, 743
 - 7.5.4 Peer-to-Peer Networks, 748
- 7.6 SUMMARY, 757

8 NETWORK SECURITY

763

- 8.1 CRYPTOGRAPHY, 766
 - 8.1.1 Introduction to Cryptography, 767
 - 8.1.2 Substitution Ciphers, 769
 - 8.1.3 Transposition Ciphers, 771
 - 8.1.4 One-Time Pads, 772
 - 8.1.5 Two Fundamental Cryptographic Principles, 776
- 8.2 SYMMETRIC-KEY ALGORITHMS, 778
 - 8.2.1 DES—The Data Encryption Standard, 780
 - 8.2.2 AES—The Advanced Encryption Standard, 783
 - 8.2.3 Cipher Modes, 787
 - 8.2.4 Other Ciphers, 792
 - 8.2.5 Cryptanalysis, 792

- 8.3 PUBLIC-KEY ALGORITHMS, 793
 - 8.3.1 RSA, 794
 - 8.3.2 Other Public-Key Algorithms, 796
- 8.4 DIGITAL SIGNATURES, 797
 - 8.4.1 Symmetric-Key Signatures, 798
 - 8.4.2 Public-Key Signatures, 799
 - 8.4.3 Message Digests, 800
 - 8.4.4 The Birthday Attack, 804
- 8.5 MANAGEMENT OF PUBLIC KEYS, 806
 - 8.5.1 Certificates, 807
 - 8.5.2 X.509, 809
 - 8.5.3 Public Key Infrastructures, 810
- 8.6 COMMUNICATION SECURITY, 813
 - 8.6.1 IPsec, 814
 - 8.6.2 Firewalls, 818
 - 8.6.3 Virtual Private Networks, 821
 - 8.6.4 Wireless Security, 822
- 8.7 AUTHENTICATION PROTOCOLS, 827
 - 8.7.1 Authentication Based on a Shared Secret Key, 828
 - 8.7.2 Establishing a Shared Key: The Diffie-Hellman Key Exchange, 833
 - 8.7.3 Authentication Using a Key Distribution Center, 835
 - 8.7.4 Authentication Using Kerberos, 838
 - 8.7.5 Authentication Using Public-Key Cryptography, 840
- 8.8 EMAIL SECURITY*, 841
 - 8.8.1 PGP—Pretty Good Privacy, 842
 - 8.8.2 S/MIME, 846
- 8.9 WEB SECURITY, 846
 - 8.9.1 Threats, 847
 - 8.9.2 Secure Naming, 848
 - 8.9.3 SSL—The Secure Sockets Layer, 853
 - 8.9.4 Mobile Code Security, 857
- 8.10 SOCIAL ISSUES, 860
 - 8.10.1 Privacy, 860
 - 8.10.2 Freedom of Speech, 863
 - 8.10.3 Copyright, 867
- 8.11 SUMMARY, 869

9 READING LIST AND BIBLIOGRAPHY 877

9.1 SUGGESTIONS FOR FURTHER READING*, 877

- 9.1.1 Introduction and General Works, 878
- 9.1.2 The Physical Layer, 879
- 9.1.3 The Data Link Layer, 880
- 9.1.4 The Medium Access Control Sublayer, 880
- 9.1.5 The Network Layer, 881
- 9.1.6 The Transport Layer, 882
- 9.1.7 The Application Layer, 882
- 9.1.8 Network Security, 883

9.2 ALPHABETICAL BIBLIOGRAPHY*, 884

INDEX 905

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PREFACE

This book is now in its fifth edition. Each edition has corresponded to a different phase in the way computer networks were used. When the first edition appeared in 1980, networks were an academic curiosity. When the second edition appeared in 1988, networks were used by universities and large businesses. When the third edition appeared in 1996, computer networks, especially the Internet, had become a daily reality for millions of people. By the fourth edition, in 2003, wireless networks and mobile computers had become commonplace for accessing the Web and the Internet. Now, in the fifth edition, networks are about content distribution (especially videos using CDNs and peer-to-peer networks) and mobile phones are small computers on the Internet.

New in the Fifth Edition

Among the many changes in this book, the most important one is the addition of Prof. David J. Wetherall as a co-author. David brings a rich background in networking, having cut his teeth designing metropolitan-area networks more than 20 years ago. He has worked with the Internet and wireless networks ever since and is a professor at the University of Washington, where he has been teaching and doing research on computer networks and related topics for the past decade.

Of course, the book also has many changes to keep up with the: ever-changing world of computer networks. Among these are revised and new material on

- Wireless networks (802.12 and 802.16)
- The 3G networks used by smart phones
- RFID and sensor networks
- Content distribution using CDNs
- Peer-to-peer networks
- Real-time media (from stored, streaming, and live sources)
- Internet telephony (voice over IP)
- Delay-tolerant networks

A more detailed chapter-by-chapter list follows.

Chapter 1 has the same introductory function as in the fourth edition, but the contents have been revised and brought up to date. The Internet, mobile phone networks, 802.11, and RFID and sensor networks are discussed as examples of computer networks. Material on the original Ethernet—with its vampire taps—has been removed, along with the material on ATM.

Chapter 2, which covers the physical layer, has expanded coverage of digital modulation (including OFDM as widely used in wireless networks) and 3G networks (based on CDMA). New technologies are discussed, including Fiber to the Home and power-line networking.

Chapter 3, on point-to-point links, has been improved in two ways. The material on codes for error detection and correction has been updated, and also includes a brief description of the modern codes that are important in practice (e.g., convolutional and LDPC codes). The examples of protocols now use Packet over SONET and ADSL. Sadly, the material on protocol verification has been removed as it is little used.

In Chapter 4, on the MAC sublayer, the principles are timeless but the technologies have changed. Sections on the example networks have been redone accordingly, including gigabit Ethernet, 802.11, 802.16, Bluetooth, and RFID. Also updated is the coverage of LAN switching, including VLANs.

Chapter 5, on the network layer, covers the same ground as in the fourth edition. The revisions have been to update material and add depth, particularly for quality of service (relevant for real-time media) and internetworking. The sections on BGP, OSPF and CIDR have been expanded, as has the treatment of multicast routing. Anycast routing is now included.

Chapter 6, on the transport layer, has had material added, revised, and removed. New material describes delay-tolerant networking and congestion control in general. The revised material updates and expands the coverage of TCP congestion control. The material removed described connection-oriented network layers, something rarely seen any more.

Chapter 7, on applications, has also been updated and enlarged. While material on DNS and email is similar to that in the fourth edition, in the past few years there have been many developments in the use of the Web, streaming media and content delivery. Accordingly, sections on the Web and streaming media have been brought up to date. A new section covers content distribution, including CDNs and peer-to-peer networks.

Chapter 8, on security, still covers both symmetric and public-key cryptography for confidentiality and authenticity. Material on the techniques used in practice, including firewalls and VPNs, has been updated, with new material on 802.11 security and Kerberos V5 added.

Chapter 9 contains a renewed list of suggested readings and a comprehensive bibliography of over 300 citations to the current literature. More than half of these are to papers and books written in 2000 or later, and the rest are citations to classic papers.

List of Acronyms

Computer books are full of acronyms. This one is no exception. By the time you are finished reading this one, the following should ring a bell: ADSL, AES, AJAX, AODV, AP, ARP, ARQ, AS, BGP, BOC, CDMA, CDN, CGI, CIDR, CRL, CSMA, CSS, DCT, DES, DHCP, DHT, DIFS, DMCA, DMT, DMZ, DNS, DOCSIS, DOM, DSLAM, DTN, FCFS, FDD, FDDI, FDM, FEC, FIFO, FSK, FTP, GPRS, GSM, HDTV, HFC, HMAC, HTTP, IAB, ICANN, ICMP, IDEA, IETF, IMAP, IMP, IP, IPTV, IRTF, ISO, ISP, ITU, JPEG, JSP, JVM, LAN, LATA, LEC, LEO, LLC, LSR, LTE, MAN, MFJ, MIME, MPEG, MPLS, MSC, MTSO, MTU, NAP, NAT, NRZ, NSAP, OFDM, OSI, OSPF, PAWS, PCM, PGP, PIM, PKI, POP, POTS, PPP, PSTN, QAM, QPSK, RED, RFC, RFID, RPC, RSA, RTSP, SHA, SIP, SMTP, SNR, SOAP, SONET, SPE, SSL, TCP, TDD, TDM, TSAP, UDP, UMTS, URL, VLAN, VSAT, WAN, WDM, and XML. But don't worry. Each will appear in **boldface type** and be carefully defined before it is used. As a fun test, see how many you can identify *before* reading the book, write the number in the margin, then try again *after* reading the book.

How to Use the Book

To help instructors use this book as a text for courses ranging in length from quarters to semesters, we have structured the chapters into core and optional material. The sections marked with a “*” in the table of contents are the optional ones. If a major section (e.g., 2.7) is so marked, all of its subsections are optional. They provide material on network technologies that is useful but can be omitted from a short course without loss of continuity. Of course, students should be encouraged to read those sections as well, to the extent they have time, as all the material is up to date and of value.

Instructors' Resource Materials

The following protected instructors' resource materials are available on the publisher's Web site at www.pearsonhighered.com/tanenbaum. For a username and password, please contact your local Pearson representative.

- Solutions manual
- PowerPoint lecture slides

Students' Resource Materials

Resources for students are available through the open-access Companion Web site link on www.pearsonhighered.com/tanenbaum, including

- Web resources, links to tutorials, organizations, FAQs, and more
- Figures, tables, and programs from the book
- Steganography demo
- Protocol simulators